

Andrew Nash

Staff Autonomous Driving Engineer | Perception | Localization | Planning

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Summary. Staff Autonomous Driving Engineer who shipped Canada’s first medium-speed SAE L4 shuttle, Transport-Canada-authorized. Full-stack AV and robotics: machine-learning perception, factor-graph localization, real-time motion planning. Tech lead and code owner of the production stack.

Experience

Aurrigo

Staff ADS Engineer

-  Oct 2025 - Present
-  Ottawa, Canada
- **Tech lead of 6 engineers** across navigation, perception, localization, and planning; code owner of two \$1M+ production deployments.
 - **Built navigation, perception, localization, and calibration** for Canada’s first medium-speed SAE L4 shuttle; Transport Canada-authorized [\[press\]](#).
 - **Shipped PyTorch camera+LiDAR 3D detection + tracking at 83.4 mAP @ 20 Hz**; classical safety channel for ISO 26262 diverse redundancy.
 - **Built real-time ROS C++ Frenet motion planner** (replacing legacy PID): 400+ parallel trajectories at 50 Hz, jerk-optimal selection against perception.
 - **Cut lateral error 10x (0.5 m to 0.05 m), more than doubled top speed (15 to 35 km/h)**: custom GTSAM factors with dynamic GPS/lidar/IMU sensor fusion; unlocked tunnel + GPS-denied terminals.
 - **Authored Aurrigo’s end-to-end Python labelling data pipeline**: annotation tool + auto-labeller lifting 2D foundation models to 3D LiDAR.

ROS C++ PyTorch CUDA TensorRT Git Computer Vision

Aurrigo

ADS Team Lead, North America

-  Oct 2023 - Oct 2025
-  Ottawa, Canada
- **Led Aurrigo’s first AV simulator from scratch**: 1000s of scenarios proving airport-customer safety; 50x speedup vs CARLA, faster-than-real-time.
 - **Solo-authored Aurrigo’s C++ GTSAM iSAM2 SLAM and mapping** (glim-based): GPU factor graph on gtsam_points, GPS-aligned LiDAR maps, all sensors.
 - **Doubled top autonomous speed (25 to 50+ km/h)** via 27x CUDA VGICP scan-match speedup, multi-LiDAR at 20 Hz, and custom Ackermann + crab-steer kinematic factors, on the same embedded hardware.

ROS C++ Python GTSAM UE4 CI/CD Linux Bash

Internships

Nasdaq (Verafin) | Machine Learning Engineer May - Aug 2021
Lifted gradient-boosting fraud-detection model precision by 6 pp at fixed recall.

Python Pandas Scikit-Learn XGBoost LightGBM AWS

Bank of Canada | Data Science Developer Sept 2020 - Apr 2021
Built anomaly-detection models automating economist data-review workflows.

Python Pandas Scikit-Learn XGBoost JavaScript Tableau

IBM | NLP Developer Jan - Apr 2020
Improved enterprise semantic matching by 8% via weighted word embeddings.

Python NLP Keras NLTK Java JavaScript Kubernetes Cognos

Royal Bank of Canada | Full Stack Data Scientist May - Aug 2019
Built full-stack semantic search prototype with NLP backend and web frontend.

Python NLP BERT Flask React JavaScript Docker

BlackBerry QNX | NLP Developer Sept - Dec 2018
Applied NLP classifiers to automate software-license tagging across QNX repos.

Python NLP Keras FastText QNX Docker MongoDB

Education

MSc Computer Science (Thesis)

Memorial University Of Newfoundland

 2021 - 2023  NL, Canada
4.0 GPA. Thesis: [Herds Eye View](#) (AIIDE-2023).

BEng Computer Engineering (Co-Op)

Memorial University Of Newfoundland

 2016 - 2021  NL, Canada
Capstone: [SR AUV](#) (best 2021 capstone).

Publications

Herds Eye View

Accepted to AIIDE-2023

 [arXiv](#)  [Blog Post](#)

Unity RL Transformer PyTorch

Projects

Sage | Real-time D&D AI Co-pilot

- Streaming audio to LLM agent at 1s end-to-end latency: Deepgram STT, Gemini agent loop, 20+ MCP tools.
- Solo-architected: 98 REST routes, hybrid RAG, Svelte 5 PWA, Docker.

FastAPI LLM RAG MCP GCP JS

IGVC | Autonomous Vehicle Competition

 [Blog Post](#)

- Trained custom transformer BEV prediction models from Unreal Engine data.
- Design placed 2nd internationally 2023.

ROS Unreal Engine C++ PyTorch

SR AUV | Autonomous Underwater Vehicle

 [Blog Post](#)

- Real-world tested end-to-end flying autonomously using PPO RL.
- Awarded best 2021 Capstone Project.

Unity C# TensorFlow TPU RL

SpaceX Hyperloop | Pod Competition

- Placed 8th at 2019 Pod Competition; built full-stack pod telemetry.

C++ Flask JS Protobuf CAN